

BACHELOR'S DEGREE (C.B.C.S) EXAMINATION, MARCH 2025
2024 ADMISSIONS REGULAR
SEMESTER II - MAJOR (COMPUTER APPLICATIONS)
CA2DSCA01B24 - Python Programming for Data Science

Time : 1.5 Hours

Maximum Marks : 50

Part A

(1x10=10)

Answer all questions. Each question carries 1 mark.

1. Analyse the issue in this comparison, if $10=2$:
2. Name one popular IDE used for Python development.
3. Recall the extension of Jupyter Notebook Files
4. Define a DataFrame
5. Interpret the relevance of 'Inplace' parameter in pandas functions
6. Identify the use of df.info() function.
7. Identify the function for creating a bar chart
8. Define Probability Distribution
9. Which type of Sampling method is Cluster Sampling.
10. Name the distribution where the weight of newborn babies follow

[CO1,Analyse]
 [CO1,Remember]
 [CO1,Remember]
 [CO2,Remember]
 [CO2,Analyse]
 [CO2,Analyse]
 [CO3,Understand]
 [CO4,Understand]
 [CO4,Apply]
 [CO4,Create]

Part B

Answer any 2 questions from the bunch of CO1. Each question carries 5 mark.

(5x2=10)

11. Compare and Contrast List and Tuple.
12. Identify the output of the following code? Discuss

```
x = 5
if x > 3:
    print("A")
elif x > 2:
    print("B")
else:
    print("C")
```

[CO1,Remember]

[CO1,Evaluate]

13. Compare and Contrast mutable and immutable data structures in python.

[CO1,Understand]

Part B

Answer any 2 questions from the bunch of CO2. Each question carries 5 mark.

(5x2=10)

14. Differentiate between NumPy Array and Pandas DataFrame
15. Explain in detail NumPy
16. Create a DataFrame df with columns "Name", "Age" and "Marks". Write a Python program to add a new column "Grade"

[CO2,Understand]

[CO2,Remember]

[CO2,Create]

Part B

Answer any 2 questions from the bunch of CO4. Each question carries 5 mark.

(5x2=10)

17. Explain the difference between simple random sampling and stratified sampling.
18. Compare and Contrast between Binomial and Poisson distributions
19. Find the mean, median and mode of the following set of data 100,112,116,130,145

[CO4,Evaluate]

[CO4,Analyse]

[CO4,Apply]